NIRAV SHAH

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SUMMARY

Software engineer with 4+ years building high scale distributed systems and REST APIs in Python and SQL on GCP and Azure. At Yahoo, re-architected core event processing to cut cloud cost 94% ($150K+/month), ran 50+ production pipelines processing 5B+ events/day at a 99.9% SLA, and was promoted to SDE II within 2 years.

EXPERIENCE

Yahoo (Mountain View, CA)

Software Development Engineer II Mar 2026 - Present

- Re-architected 5 high volume event pipelines processing from Pig on Dataproc into 3 partitioned BigQuery SQL jobs, cutting infrastructure cost 94% ($161K to $9K/month) and runtime 85%; authored the design doc.
- Built a near real-time analytics pipeline in Python, SQL and Airflow ingesting 4M views/hour at 10-minute freshness for $1.2K/month, used by 100+ dashboard users across 10+ Yahoo teams.
- Debugged a production data quality defect inflating engagement rate 28% for Yahoo News, Homepage and Sports, tracing 884K phantom sessions to event logic and filtering them out in the ETL.
- Built an AI agent workflow (Claude) that automates schema changes, cutting each change from 2 weeks to 2 days and shipping 20+ metrics, including users, impressions, and sessions.
- Restored content attributes for 85% of real-time hourly views after an upstream source was retired; built, deployed, and validated an Apache Beam Dataflow fix that permanently moved enrichment to the new source.

Software Development Engineer I Feb 2024 - Feb 2026

- Led a team of 8 engineers migrating batch workflows from on-prem Oozie and Kubernetes to GCP Cloud Composer, redesigning 10+ DAGs in Python with unit tests and cutting peak runtime 20%.
- Maintained 50+ batch and streaming pipelines (Pub/Sub, Kafka, Storm; 5B+ events/day) at 99.9% SLA as on-call owner, integrating OpsGenie alerting with Airflow and cutting incident response time 50%.
- Engineered an ML (ARIMA) anomaly detection model across 30 content and video KPIs, catching 10+ anomalies in 3 months, including a URL change misrouting homepage views, fixed in 5 hours.
- Redesigned referral classification logic for a certified audience-measurement pipeline, reducing uncategorized pageviews 92% (20M to 1.5M/day) and improving downstream reporting accuracy.

Software Development Engineer Intern May 2023 - Aug 2023

- Built a BigQuery query benchmarking tool by using a REST API to label BigQuery jobs, testing user queries behind 7+ API endpoints and flagging 50 slow, resource-intensive queries for optimization.
- Extended an internal metrics REST API to auto label jobs with a UUID, date window and range label, built a UI to benchmark multiple date ranges from centralized location, and stored runtime stats in Spanner.

PROJECTS

Expertiza: An Open-Source Peer Assessment
Built React features and refactored Ruby backend modules using object-oriented principles and design patterns.

GoodBowls: Research Assistant
Developed a Flutter based mobile app and React admin dashboard to deliver interactive culinary education.

CityByte:
Built a Python/Django web app for location insights; added multi-threading and caching to cut load times 50%.

EDUCATION

North Carolina State University (Master of Computer Science) **GPA: 4.0/4.0** Aug 2022 - Dec 2023
Pune Institute of Computer Technology (B.E. in Computer) **GPA: 8.99/10** Aug 2016 - May 2020

TECHNICAL SKILLS

Languages:	Python, SQL, Java, C, C++, JavaScript
Systems & Infra:	REST APIs, Kubernetes, Docker, Terraform, CI/CD (Jenkins, GitHub Actions), Pub/Sub
Data Systems:	BigQuery, Spark, Apache Beam (Dataflow), Airflow, Databricks, Delta Lake, Spanner
Monitoring & BI:	Looker, Tableau, Power BI, Rootly, BigPanda, OpsGenie, pytest